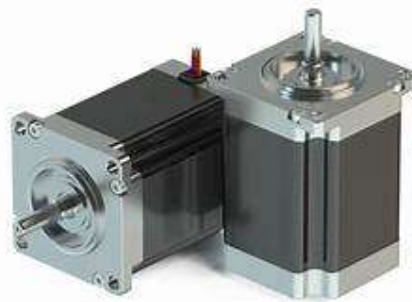


Servo Motors – Working Principle, Control Signals, and Applications

Introduction

Servo motors are a special type of rotary actuator that allow precise control of angular position, and in some cases, velocity and acceleration. They are widely used in robotics, automation, aerospace, and consumer electronics due to their compact size, high torque, and accuracy. Unlike regular DC motors, servo motors are capable of moving to a specific angle and holding that position, making them ideal for position control tasks such as steering, robotic arm movement, and camera stabilization.



Working Principle of Servo Motors

A servo motor functions as part of a closed-loop control system. This means the motor's output (position) is continuously monitored and compared with the desired input, and any difference (error) is corrected in real time. A typical servo motor consists of the following key components:

1. DC Motor or AC Motor – The actual motor that produces rotational movement.
2. Gearbox – Reduces the motor's speed and increases torque, enabling precise control.
3. Control Circuitry – An embedded electronic circuit that receives control signals and regulates motor operation.
4. Position Sensor (Potentiometer or Encoder) – Measures the shaft's position and feeds it back to the control circuit.

The working process of a servo motor starts when a control signal (usually a pulse-width modulated signal) is sent to the motor's input. The control circuit interprets this signal and compares the desired position with the current position detected by the position sensor. If the current position differs from the desired one, the motor is rotated in the necessary direction until the error is minimized, i.e., until the motor reaches the correct

position. Once it reaches that position, the motor stops and holds there unless a new signal is received.

Control Signals

Servo motors are typically controlled by PWM (Pulse Width Modulation) signals. Although they use PWM, it is not used for speed control (as with DC motors), but rather for position control.

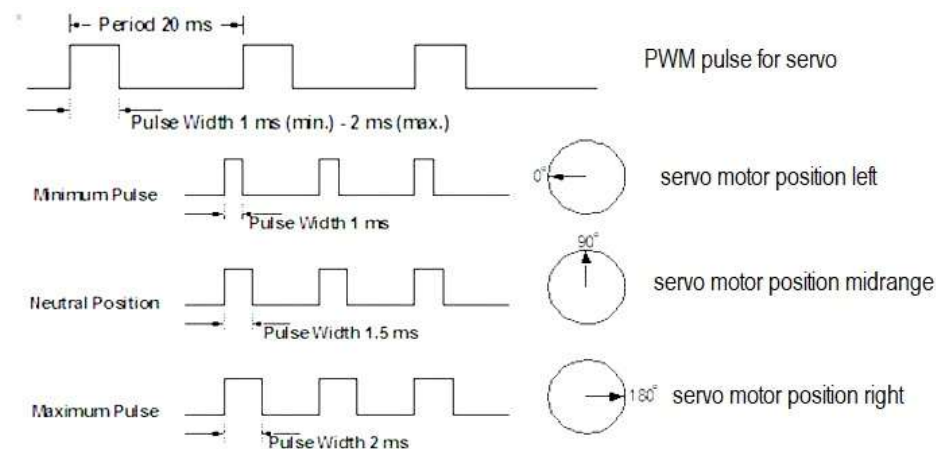
- The control signal is a series of square pulses.
- The pulse width (duration the signal stays HIGH) determines the angle to which the motor shaft should turn.
- A typical servo motor expects a pulse every 20 milliseconds (ms), i.e., 50 Hz frequency.

Pulse Width and Corresponding Angle:

- 1 ms pulse width $\rightarrow$ 0° (minimum angle)
- 1.5 ms pulse width $\rightarrow$ 90° (midpoint)
- 2 ms pulse width $\rightarrow$ 180° (maximum angle)

The range of movement varies between different servos. Commonly, hobby servos rotate 0° to 180°, but some servos can rotate up to 270°, and continuous rotation servos can rotate indefinitely like DC motors.

The PWM signal is usually generated by a microcontroller (e.g., Arduino, Raspberry Pi, or any embedded system). The servo motor's control pin reads the PWM signal, while the power and ground pins supply energy to operate the motor.



Types of Servo Motors

1. Positional Rotation Servo – Most common, rotates within a fixed range (usually 0°–180° or 0°–270°).
2. Continuous Rotation Servo – Rotates continuously in either direction, controlled by PWM signal for speed and direction.
3. Linear Servo – Converts rotational motion into linear motion, used where straight-line movement is needed.

Applications of Servo Motors

Servo motors are used in diverse fields due to their accuracy, control, and compact design. Here are some of their most significant applications:

- **Robotics:** Servo motors are crucial in robotic arms, legs, and grippers, where precise motion and holding position are essential. They allow robots to perform repetitive and complex movements with consistency.
- **RC (Remote-Controlled) Devices:** In RC cars, boats, airplanes, and drones, servos control steering mechanisms, wing flaps, and camera gimbals, ensuring smooth and responsive actions.
- **3D Printers:** Servo motors ensure precise control over the print head and platform, improving print quality and consistency.
- **Camera Systems:** Pan-tilt camera mounts use servos to adjust camera angles smoothly, essential in surveillance and videography.

Advantages of Servo Motors

- **Precise Position Control:** Ability to move to a specific angle and hold it.
- **High Torque at Low Speeds:** Gearbox enhances torque output.
- **Compact Size:** Ideal for space-constrained applications.
- **Energy Efficient:** Draws power only when movement is needed.
- **Feedback Mechanism:** Ensures accurate, closed-loop control.